



A smart obstacle avoiding technology based on depth camera for blind and visually impaired people

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Abstract

It remains challenging to assist BVI individuals in outdoor travel nowadays. In this paper, we propose a set of low-cost wearable obstacle avoidance devices and introduce an obstacle detection algorithm called L-PointPillars, which is based on point cloud data and is suitable for edge devices. We first model the obstacles faced by BVI individuals during outdoor travel and then establish a mapping between the information space and physical space based on point clouds. We then introduce depthwise separable convolution and attention mechanisms to develop L-PointPillars, a fast neural network for obstacle detection. This network is specifically designed for creating wearable obstacle detection devices. Finally, we implemented a wearable electronic travel aid device (WETAD) based on L-PointPillars on the Jetson Xavier NX. Experiments show that while L-PointPillars reduces the number of parameters in the original PointPillars by 75%, WETAD achieves an average obstacle detection accuracy of 95.3%. It takes an average of 144 milliseconds to process each frame during outdoor travel for BVI individuals, which is more than twice as fast as the Second network and 31% improvement compared to PointPillars.

Keywords Visual impairment · Travel aid · Feature extraction · Obstacle detection · Point cloud

1 Introduction

Visual impairment is a significant global health issue, affecting approximately 285 million individuals worldwide, of whom 39 million suffer from blindness, according to the World Health Organization (World Health Organization 2012). Within China, more than 17.5 million people experience some form of visual impairment, with over 5 million classified as blind (World Blind Union 2018). BVI individuals face significant

challenges when it comes to independently navigating and traveling outdoors. This is primarily due to difficulties in avoiding obstacles and identifying destinations (Sen et al. 2018). The mobility needs of BVI individuals have attracted significant research attention on assistive technologies.

Advances in sensor technology, such as laser scanners, sonar, ultrasonic systems, and camera systems, have facilitated the creation of electronic mobility aids that assist BVI individuals in detecting and navigating around obstacles. For example, Kumar et al. proposed an ultrasonic cane to identify ground and overhead barriers (Kumar et al. 2014). Ahlmark et al. developed a haptic interface and laser range finder to virtually scan the environment (Ahlmark et al. 2013). Wahab et al. developed a smart cane that utilizes ultrasonic sensors, a microcontroller, a vibrator, a buzzer, and a water detector to detect obstacles (Wahab et al. 2011). Aymaz et al. designed an ultrasonic headset to assist BVI individuals in navigating obstacles (Aymaz and Çavdar 2016). However, these canes are limited in their ability to detect obstacles within a short range and identify only a few attributes. This is because they have constrained computing power, despite being low-cost (Jiao et al. 2016).

Advances in sensor technology, such as laser scanners, sonar, ultrasonic systems, and camera systems, have

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facilitated the development of electronic mobility aids that assist BVI individuals detect and navigate around obstacles. For example, Kumar et al. proposed an ultrasonic cane to identify ground and overhead barriers (Kumar et al. 2014). Ahlmark et al. developed a haptic interface and laser range finder to virtually scan the environment (Ahlmark et al. 2013). Wahab et al. developed a smart cane that utilizes ultrasonic sensors, a microcontroller, a vibrator, a buzzer, and a water detector to detect obstacles (Wahab et al. 2011). Aymaz et al. designed an ultrasonic headset to assist BVI individuals in navigating obstacles (Aymaz and Çavdar 2016). However, these canes have limitations in detecting obstacles within a short range and can only identify a few attributes. This is because they have limited computing power, despite being low-cost (Jiao et al. 2016).

With the miniaturization of cameras and the improvement in resolution, extensive research has been conducted on vision-based techniques for obstacle avoidance. Tapu et al. developed a real-time obstacle detection system based on apparent motion analysis using the multiscale Lucas-Kanade algorithm (Tapu et al. 2014). Gharani et al. proposed a method that combines optical flow and context awareness to detect obstacles from consecutive image frames (Gharani and Karimi 2017). Costa et al. presented a 2D ensemble empirical mode decomposition image optimization and a two-layer disparity image segmentation algorithm to detect and classify obstacles (Costa et al. 2016). Lee et al. developed a robust depth-based framework for guiding users with free-moving cameras to avoid obstacles (Lee et al. 2016). Berenguel-Baeta et al. combined RGB-D and fisheye cameras to perceive the environment and navigate around obstacles (Berenguel-Baeta et al. 2020). Compared to 2D images, 3D point clouds contain rich geometric and spatial details. Object detection methods using 3D point clouds have been developed for applications such as autonomous driving and robotics. For instance, PointPillars, a promising one-stage 3D object detection technique, has achieved state-of-the-art accuracy and fast inference on the KITTI benchmark (Lang et al. 2019; Geiger et al. 2012). However, relying on high-performance graphics processing units (GPUs) presents challenges in developing affordable travel aids for BVI individuals.

Current obstacle detection algorithms heavily depend on high-performance GPUs or NPUs. For example, Second (Liu et al. 2018) accelerates 3D convolution calculations using sparse operators, while PointPillar (Lang et al. 2019) converts point clouds into pseudo-images and utilizes 2D convolutional networks for faster processing. However, edge devices typically have limited computing capabilities, making it particularly important to design obstacle detection algorithms that are suitable for such devices.

Our goal is to develop a real-time obstacle detection algorithm that can be efficiently deployed on edge devices.

Additionally, we aim to develop a range of wearable obstacle avoidance devices. First, we model the obstacles that BVI individuals encounter when traveling outdoors. We establish a correlation between the information space and physical space by utilizing point clouds. We have modified the PointPillars (Lang et al. 2019) to create a more efficient obstacle detection network called L-PointPillars. L-PointPillars is specifically designed for embedded wearable systems with limited computing resources. Finally, we implemented L-PointPillars on the NVIDIA Jetson Xavier NX to develop a travel aid device for visually impaired individuals. The main contributions are as follows:

(1) We propose Lean PointPillars, an obstacle detection algorithm designed for low computing power scenarios on edge devices. Lean PointPillars is an optimized 3D obstacle detection neural network that reduces the number of parameters in the original PointPillars (Lang et al. 2019) by 75%, while still maintaining high accuracy. This optimization enables real-time performance on embedded hardware. Compared to PointPillars (Lang et al. 2019), Lean PointPillars sacrifices a small amount of accuracy in exchange for significantly faster inference speed. This is achieved by utilizing a more efficient backbone and attention mechanism.

(2) We have developed a portable and user-friendly wearable edge device and created an innovative wearable assistive system that detects obstacles and offers real-time auditory information to BVI individuals. Lean PointPillars is deployed on the Jetson Xavier NX edge device, and the RealSense D435i camera is used to capture input point cloud data.

By addressing the technical challenges of obstacle perception and designing interfaces tailored for BVI individuals, this wearable system aims to enable safer and more independent navigation for travelers with BVI who have diverse mobility needs. The results of this study demonstrate an important initial step towards developing more intelligent assistive technologies that could ultimately enhance accessibility and quality of life.

2 Related work

As camera miniaturization and resolution increase, researchers have studied various vision-based obstacle avoidance techniques. For example, Bouhamed et al. (2013) proposed a smart cane that utilizes ultrasonic sensors and a monocular camera to detect steps. By extracting the frequent features of ultrasonic signals, the smart cane can classify steps of various shapes. Zhang et al. (2014) used a smartphone camera to collect data in front of BVI individuals. They developed the Faster R-CNN and YOLO algorithms to detect obstacles, which were then executed on the smartphone. Gharani and Karimi (2017) introduced optical flow technology to develop

a context-aware obstacle detection method. The monocular camera has the advantage of a simple structure because it uses only a single camera to collect environmental information. However, the absence of spatial 3D information regarding environmental objects makes it challenging to calculate the distance between the BVI individuals and the obstacle.

Binocular cameras, which can calculate the precise coordinates of an object based on the disparity between the two cameras, were introduced to acquire spatial information about obstacles. For example, Zhu et al. (2016) integrated a binocular camera, image processor, gyroscope, and GPS to develop a wearable undershirt. This undershirt provides vibration and voice feedback for BVI individuals as soon as it detects obstacles. Jiang et al. (2018) proposed a wearable system that captures images at a fixed frequency using binocular cameras. These captured images are then analyzed on a cloud server to assist visually impaired individuals in making more informed decisions about their next steps. Wang et al. (2018) proposed a method for obstacle avoidance based on binocular vision. In the process of obstacle detection, we first utilize an algorithm that relies on depth detection to ascertain the presence of obstacles. Then, we utilize an obstacle information acquisition algorithm based on frame difference to determine the orientation and speed of both static and dynamic obstacles. The binocular camera captures both the depth and color of an object, but it is sensitive to various factors in complex environments, such as lighting conditions, chromaticity, and shadows.

Compared to the vision sensors mentioned above, RGB-D sensors can not only capture depth data that reflects the distance of an object but also obtain color data that corresponds to the depth data (He and Liu 2022). Hence, researchers have also utilized RGB-D sensors to develop travel assistance devices for BVI individuals. For example, Orita et al. (2013) integrated an RGB-D camera, a keyboard, and a vibration motor to develop a blind cane. They utilized a 3D line fitting technique to identify obstacles on the ground and classified the space above the ground as an obstacle. Once an obstacle is detected, the vibration feedback is transmitted to the user. Aladren et al. (2014) introduced RGB-D cameras to indoor scenes and developed a system that can assist visually impaired individuals in navigating indoor obstacles. This system is based on color and depth maps captured by the RGB-D cameras. Pham et al. (2016) used Kinect to capture depth images of the environment. They then used the depth information to segment the multi-plane structures in the scene and classify the planes (e.g., obstacles, walls, and floors) based on their characteristics. Rodríguez et al. (2012) generated a point cloud based on the depth map and utilized a random sampling consistency algorithm to fit the point cloud and identify the ground. Finally, information about obstacles is obtained by excluding the ground data. However, the method has the drawback of misidentifying

flat surfaces, such as walls and tables, as the ground. Cheng et al. (2015) designed a seed region growth algorithm based on depth information for ground detection and successfully segmented the ground and obstacles.

The RGB-D sensors are highly adaptable to different environments, making them ideal for obstacle detection tasks. Currently, the most popular consumer RGB-D cameras on the market are Microsoft's Kinect, Stereolab ZED, and Intel RealSense (He and Liu 2022). Among them, Kinect is primarily used in indoor environments. ZEDs are too expensive to promote the application for visually impaired individuals. The RealSense 3D camera offers the advantages of being small in size and having high resolution. The D435i product, in particular, includes built-in inertial sensors. These sensors allow for not only capturing color and depth information of the observed object, but also obtaining data on the object's acceleration and angular velocity. It has been widely used in robotics, virtual reality, and augmented reality. Therefore, we introduce the Intel RealSense D435i in our paper as a travel aid device for visually impaired individuals.

3 Methodology

In order to accurately determine the position of obstacles during outdoor travel for BVI individuals, our paper first analysis BVI individuals' outdoor travel scenes, then models the obstacles in the outdoor travel scene using point cloud data. We then establish the mapping relationship between the obstacles in the camera coordinate system and the world coordinate system.

3.1 BVI individuals' outdoor travel scene analysis

The daily travel scenes for visually impaired individuals described in this article mainly focus on city roads. These scenes can be divided into two main categories:

- (1) Walking on blind paths or sidewalks. Most city sidewalks have blind paths, but in some cases, such as roads within residential areas, there may not be blind paths. Through on-site investigations, it was found that the characteristic of this type of scene is relatively static. This means that most objects in the scene are not moving, and any moving objects have a slow speed. Subsequent surveys of visually impaired individuals confirmed this characteristic, indicating that the majority of their travel scenes fall into the first category. The most common obstacles for visually impaired individuals in these scenes are parked bicycles, parked vehicles, and pedestrians.
- (2) Walking on the road. This includes visually impaired individuals crossing roads or occasionally walking

on narrow roads without sidewalks. Through on-site observations, it was found that the characteristic of this type of scene is highly dynamic. This means that many objects in the scene are in motion, and their speed is relatively fast. The survey results also reflect this characteristic, showing that pedestrians and cyclists move quickly when crossing the road, which may impede the normal walking of visually impaired individuals. In some cases, cars may also cross the stop line and stop on the zebra crossing, obstructing the movement of visually impaired individuals.

By analyzing the characteristics of these two types of scenes, we can identify the three most common obstacles that visually impaired individuals may encounter in their daily travel: cars, pedestrians, and cyclists.

3.2 Modeling the outdoor obstacle

According to Fig. 1, the obstacle in the BVI individuals outdoor scene could be modeled using the world coordinate system, denoted as $O_w - X_w Y_w Z_w$. The captured depth image of the obstacle could be modeled by the deep camera coordinate system, denoted as $O_c - UV$. The camera coordinate system $O_c - UV$ represents the image plane of the depth camera. Each pixel on this plane has three attributes: u for the abscissa, v for the ordinate, and d for the depth value. The world coordinate system $O_w - X_w Y_w Z_w$ describes the position of the obstacles, which are represented as a set of points. Each point has three attributes, which are the coordinates along the three axes. The origin of the two systems is aligned, and the Y_w axis is aligned with the direction in which the depth camera is facing.

The depth camera is attached to the chest of BVI individuals to capture RGB-D images of obstacles. Each pixel corresponds to a point in physical space. Then, each pixel

of the image captured by an RGB-D camera can be transformed into points described by the world coordinate system, as shown in Eq. 1.

$$\begin{cases} x_w = \frac{y_w * (u - C_x)}{f_x} \\ z_w = \frac{y_w * (u - C_y)}{f_y} \\ y_w = \frac{d}{scale\ factor} \end{cases} \quad (1)$$

Where f_x and f_y represent the focal length, and C_x and C_y denote the principal point offsets. The scale factor relates the physical dimensions of the obstacles to their corresponding dimensions in the depth image. The parameters mentioned above are intrinsic parameters of a depth camera, which can be obtained through camera calibration.

3.3 Data denoising and rectification

During outdoor travel, both the natural movement of the body and changes in environmental lighting can cause noise and distortion when an RGB-D camera captures images. The noises and distortion could affect the accuracy of obstacle detection. Hence, point cloud denoising and rectification technologies are introduced as methods used to reduce noise and distortion.

The bilateral filter is applied to denoise the point cloud data in our paper. The bilateral filter is based on the Gaussian filter, which uses a kernel to convolve the point cloud data. The value of the center point is determined by calculating the weighted average of the adjacent points in the Gaussian kernel. The weight of the kernel is determined by the distance between the central point and adjacent points. The smaller the distance, the greater the weight. Bilateral filtering not only considers the distance between adjacent points, but also improves the Gaussian filter by incorporating the correlation between the value and weight of neighboring points. For a point p , the bilateral filter can be described by Eq. 2.

$$\hat{p} = p + \lambda n_i \quad (2)$$

Where n_i represents the normal vector of point p , and λ denotes the bilateral filter factor, which is defined according to Eq. 3.

$$\lambda = \frac{\sum_{q \in s} \omega_s(\|q - p\|) \omega_r(\|\langle n_p, n_q \rangle\| - 1) \langle n_p, p - q \rangle}{\sum_{q \in s} \omega_s(\|q - p\|) \omega_r(\|\langle n_p, n_q \rangle\| - 1)} \quad (3)$$

Where s represents the neighborhood points of p , ω_s and ω_r are two Gaussian functions defined according to Eq. 4.



Fig. 1 The model of obstacles

$$\begin{cases} \omega_s(x) = e^{-\frac{x^2}{2\delta_s^2}} \\ \omega_r(x) = e^{-\frac{x^2}{2\delta_r^2}} \end{cases} \quad (4)$$

The variable ω_s represents a spatial Gaussian function that determines the weight based on the distance between the center point and the neighboring points. On the other hand, ω_r represents a range Gaussian function. δ_s and δ_r are two constants. In our paper, δ_s and δ_r are set to 0.16 and 0.125, respectively.

Since the Intel RealSense D435i depth camera also integrates an Inertial Measurement Unit (IMU), it can detect the camera's rotation angles along three axes. The rotation angles are introduced to correct the point cloud data according to Equation (5) in our paper. For a point $p(x_p, y_p, z_p)$ in the denoised point cloud data, the rectified point can be described as $p(\hat{x}_p, \hat{y}_p, \hat{z}_p)$ according to Eq. 5.

$$\begin{bmatrix} \hat{x}_p \\ \hat{y}_p \\ \hat{z}_p \end{bmatrix} = R_1 R_2 R_3 \begin{bmatrix} x_p \\ y_p \\ z_p \end{bmatrix} \quad (5)$$

In which R_1 , R_2 , and R_3 can be defined as Eq. 6.

$$\begin{cases} R_1 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos \alpha & \sin \alpha \\ 0 & -\sin \alpha & \cos \alpha \end{bmatrix} \\ R_2 = \begin{bmatrix} \cos \beta & 0 & -\sin \beta \\ 0 & 1 & 0 \\ \sin \beta & 0 & \cos \beta \end{bmatrix} \\ R_3 = \begin{bmatrix} \cos \gamma & \sin \gamma & 0 \\ -\sin \gamma & \cos \gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{cases} \quad (6)$$

Which (α, β, γ) are the rotation angles along the x , y , and z axes.

3.4 Performance metrics

The average precision (AP) is used to measure the accuracy of the detection. Different difficulty levels of obstacles need to be compared separately in terms of AP. For the performance of the detection, the number of samples processed per second (fps) is used as a measure. The calculation method of AP is defined by Eqs. 7:

$$precision = \frac{tp}{tp + fp} = \frac{tp}{n} \quad (7)$$

In the formula *precision* represents the correct parts detected by the network, *tp* represents the correctly detected parts, *fp*

represents the incorrectly detected parts, and *n* represents the sum of the two, means all the detections.

Recall represents the proportion of true targets detected by the model, defined by Eqs. 8:

$$recall = \frac{tp}{tp + fn} \quad (8)$$

In the formula, *fn* represents the parts that the model did not detect.

The Precision-Recall (PR) curve plotted based on precision and recall can well represent the detection performance of the network model. AP is a numerical value calculated based on the PR curve, which can represent the PR curve and intuitively reflect the situation of the PR curve. The calculation method using 11 equally spaced recall values, taking the average of the maximum precision values in each recall interval as the AP of the classifier, and AP can be defined by Eqs. 9. In the formula, $P(r)$ represents the precision when the recall is r .

$$AP = \frac{1}{11} \sum_{r \in (0, 0.1, \dots, 1)} \max_{\hat{r}: \hat{r} > r} P(r) \quad (9)$$

Meanwhile, the obstacle detection algorithm based on L-PointPillars processes each frame and calculates the distances of the obstacles. The detection results of each scene were recorded in the log file so that we can compare them to the actual scenes.

4 The architecture of the l-pointpillars

We modified the PointPillars to develop a fast obstacle detection network called L-PointPillars, and Fig. 2 illustrates its architecture. L-PointPillars is primarily composed of a modified Pillar Feature Extraction Network, a backbone, and a detection head. In this paper, we employ point cloud data encoding technology for point cloud encoding. Firstly, according to the concept of voxelization, the original point cloud is divided into blocks from a bird's-eye view perspective. The internal features of the point cloud blocks are initially extracted using a multi-layer perceptron. Then, the features of the point cloud blocks are discretized to their respective initial positions in order to preserve the relative spatial features between the blocks. The discretized point cloud data is encoded as a regular, multi-channel, two-dimensional image. Secondly, we build upon the region proposal network and introduce the concept of depth separable convolution. A network is designed to extract low-complexity obstacle features. This network performs multiple downsampling operations on the multi-channel two-dimensional image and uses transpose convolution to

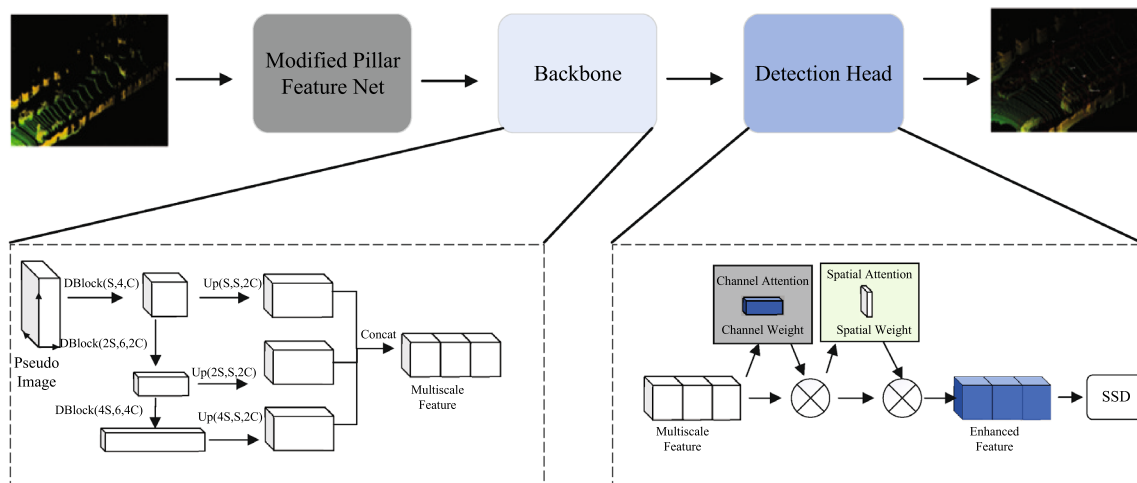


Fig. 2 The network architecture of L-pointPillars

upsample the feature maps obtained from downsampling. The upsampled features are then concatenated. The purpose of feature map concatenation is to merge obstacle features of different scales in order to achieve improved detection results. Finally, we combine SSD with an attention mechanism. The spatial features of the obstacles are extracted and then transformed into feature maps of various sizes using convolutional layers. These feature maps are then passed to the classification prediction layer, which generates multiple candidate boxes and their corresponding class probabilities. Finally, non-maximum suppression is performed to obtain the final prediction results.

4.1 Modified pillar feature net

Modified Pillar Feature Net serves as a feature encoder, which accepts point clouds as input and converts each point cloud to a sparse pseud. As shown in Fig. 3, the point cloud is first divided in the X–Y plane into equal squares, forming a set of 3-dimensional cells known as pillars. The points in each of the pillars are then extended to a 9-dimensional space by adding the coordinate offsets between the points, the arithmetic center of the pillar, and the distances between

the points and the geometric center of the pillar. The pillars are arranged in the shape of a dense tensor with dimensions (D, P, N), where D is equal to 9, P represents the number of non-empty per sample, and N represents the number of points per pillar. Since the simplified PointNet used in PointPillar could not fully extract spatial features from each pillar. We introduce a multi-layer perceptron (MLP) for extracting per-point features in each pillar. The MLP consists of three hidden layers with neuron sizes of 16, 32, and 64, respectively. Each hidden layer is followed by batch normalization and the Mish activation function (Misra 2019). The MLP transforms the stacked pillars into a tensor, specifically per-point features, with a size of (C, P, N). Then, a max operation is performed over the N-axis to extract the most prominent features of each pillar, creating a tensor called per-pillar features with a size of (C, P). Finally, the per-pillar features are reassembled into the original pillar locations to create a tensor, known as a pseudo image, with a size of (C, H, W), where H and W represent the height and width of the canvas, respectively. Once encoded, a 2D convolution can be applied to the pseudo image.

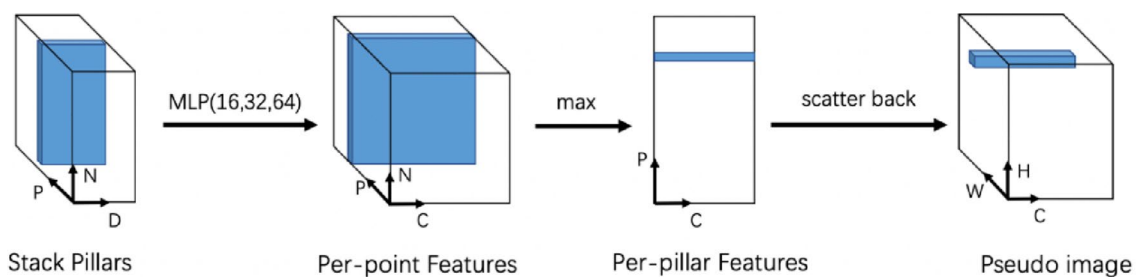


Fig. 3 Modified pillar feature net

4.2 Backbone

We introduce depthwise separable convolution, which consists of a depthwise convolution and a 1×1 pointwise convolution, to construct a backbone. In depthwise convolution, each input channel is convolved separately. Convolution is applied to one channel at a time, unlike standard convolution where it is applied to all channels simultaneously. Additionally, a 1×1 convolution is applied to the channels, linearly combining all the outputs of the depthwise convolutions. Depthwise separable convolutions have fewer parameters than standard convolutions and require much less computational cost.

As shown in Fig. 4, we employ a similar backbone to PointPillar, which consists of a down-sample network and an up-sample network. The down-sample network consists of a series of depthwise separable blocks, namely DBlock, with a size of (S, L, F) . Each DBlock operates at stride of S and consists of L 3×3 2D convolutional layers with F output channels. And each DBlock is followed by BatchNorm and ReLU, respectively. The up-sampling network takes the feature maps of each DBlock as input and performs up-sampling from an input stride of S_{in} to an output stride of S_{out} with F output channels using transposed convolution

$Up(S_{in}, S_{out}, F)$. Next, batch normalization and the Mish activation function are used to the up-sampled features. The final output of the backbone is the concatenation of all up-sampled features originating from different strides.

The pseudo image from Pillar Feature Net has 64 output channels. The down-sample network consists of 3 DBlocks: $DBlock1(S, 4, C)$, $DBlock2(2S, 6, 2C)$, and $DBlock3(4S, 6, 4C)$, where S is set to 2. Each DBlock has a corresponding Up block: $Up1(S, S, 2C)$, $Up2(2S, S, 2C)$, and $Up3(4S, S, 2C)$.

4.3 Detection head

SSD (Liu et al. 2018) is introduced to build the detect for detecting and estimating the 3D boxes surrounding the objects. The objects are detected on a 2D grid, and the position of each object along the Z-axis is an additional regression target. 2D Intersection over Union (IoU) is used to match the anchors with the ground truth. Additionally, an attention module is introduced before the SSD to enhance the multi-scale feature map and improve detection performance. As shown in Fig. 5a, the multiscale feature map is first fed into the channel attention submodule to enhance the features of a specific channel. Then, a spatial attention

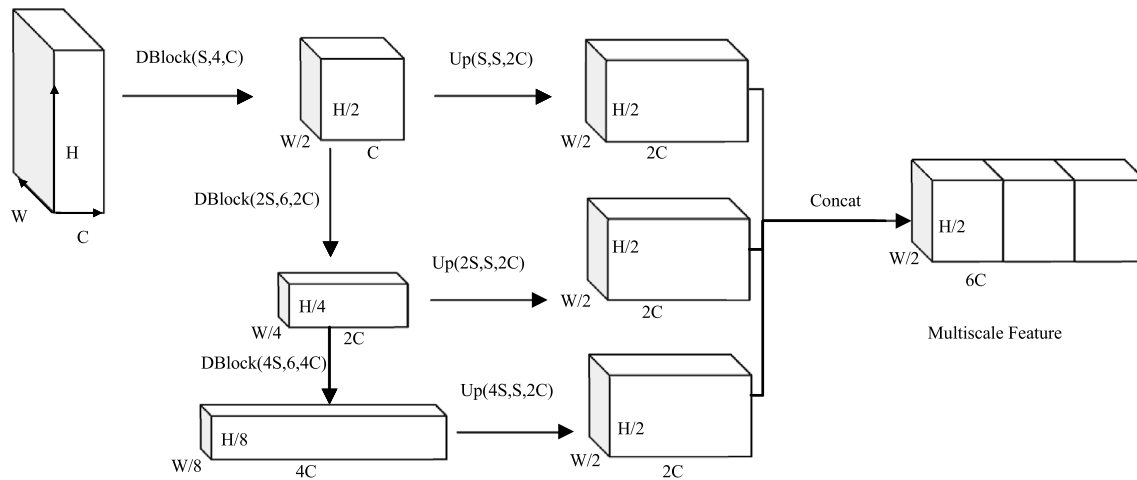


Fig. 4 Backbone based on depthwise separable Convolution

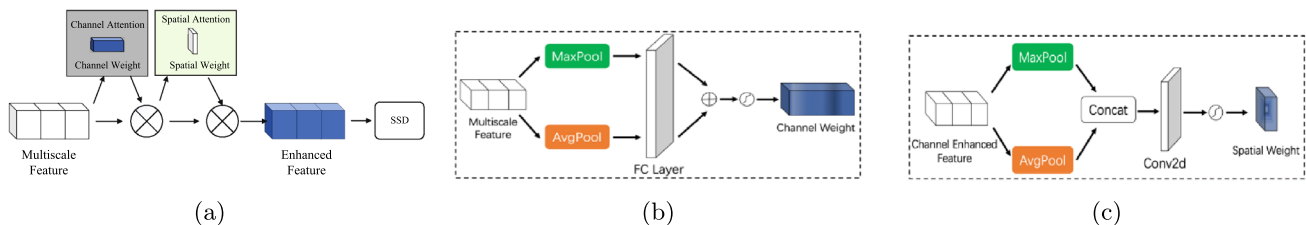


Fig. 5 Attention model **a** The architecture of attention module, **b** Illustration of channel attention submodule, **c** Illustration of spatial attention submodule. $\otimes$ denotes elementwise multiplication, $\oplus$ denotes elementwise summation, σ denotes sigmoid activation function

submodule is used to emphasize the important regions of the feature map. At last, the enhanced feature map is fed into the SSD.

As shown in Fig. 5b, max pooling is used to reduce the spatial dimension, while average pooling is used to aggregate spatial features. Then, the two aggregated feature maps are processed by a fully connected layer and merged through element-wise summation. Finally, the merged feature is passed through the ReLU activation function to generate channel weights. The enhanced channel feature is calculated according to Eqs. 10 and 11.

$$W_{channel} = \text{Sigmoid}(FC(\text{MaxPool}(F)) + FC(\text{AvgPool}(F))) \quad (10)$$

$$F_{channel} = W_{channel} \cdot F \quad (11)$$

Where F represents the multiscale feature map produced by the backbone, $W_{channel}$ denotes the scalar channel attention weight, and $F_{channel}$ refers to the channel-enhanced feature map.

The spatial attention submodule is shown in Fig. 5c. Max pooling and average pooling are applied along the channel axis to aggregate channel features. Then, the two spatial feature maps are concatenated and fed into a convolutional layer to generate spatial attention weights. The enhanced feature map is calculated according to Eqs. 12 and 13.

$$W_{spatial} = \text{Sigmoid}(\text{Conv2d}(\text{MaxPool}(F_{channel}) \oplus \text{AvgPool}(F_{channel}))) \quad (12)$$

$$F_{enhanced} = W_{spatial} \cdot F_{channel} \quad (13)$$

Where $\oplus$ represents vector concatenation, $W_{spatial}$ represents spatial attention weight, and $F_{enhanced}$ represents the enhanced feature map.

5 Implementation

We have developed a wearable electronic travel aid device (namely WETAD) to assist BVI individual avoid obstacles during outdoor environments and avoiding obstacles. The WETAD consists of the Nvidia Jetson Xavier NX, an RGB-D camera, a speaker, and software that incorporates L-PointPillar for obstacle detection. It could provide audio feedback to BVI individuals as soon as it detects an obstacle.

5.1 Hardware implementation

The Nvidia Jetson Xavier NX module packs the power of the NVIDIA Xavier SoC into a compact size of 70 mm x 45 mm. Since it has an NVIDIA Volta architecture GPU with 384 NVIDIA CUDA cores and a 6-core NVIDIA Carmel ARM CPU, it brings incredible new capabilities to small and power-efficient AI systems. Hence, the Nvidia Jetson Xavier NX is selected as the core processing module for developing the wearable travel aid devices, as shown in Fig. 6a. The Intel RealSense D435i RGB-D camera is integrated with the Nvidia Jetson Xavier NX through a USB interface. It captures a depth image and then transforms it into point cloud data based on the method described in section 3. Two speakerphones are connected to an external USB sound card, which is integrated with the Jetson Xavier NX through a USB interface. The WETAD could provide auditory information about the types and distances of obstacles ahead to BVI individuals through speakerphones. A regular portable power bank, capable of outputting 5V at 3A, is used as the power source. The power bank is connected to the Jetson Xavier NX using a USB DC power plug barrel connector cable. Figure 6b shows the prototype of the WETAD.

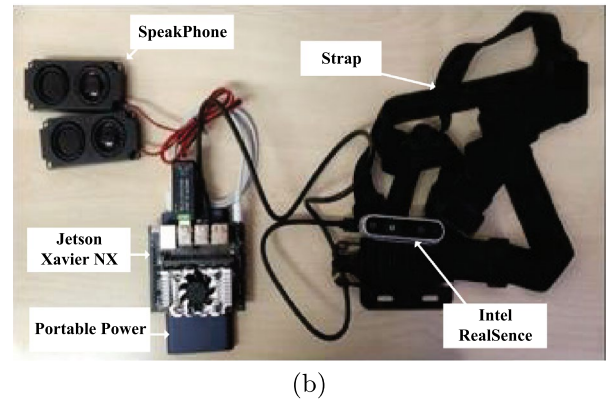
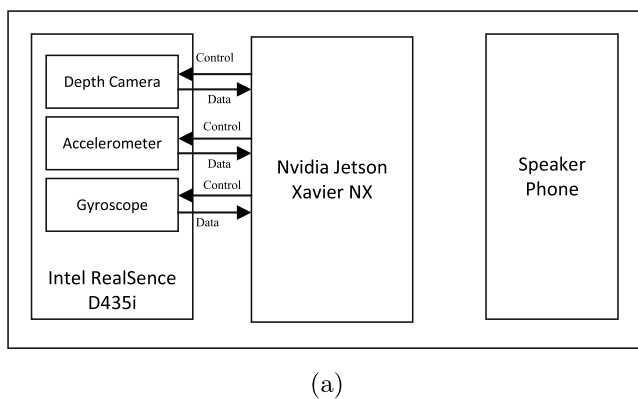


Fig. 6 The WETAD. **a** The architecture of the WETAD. **b** Physical display of the WETAD

5.2 Software implementation

The L-PointPillars is implemented using PyTorch and Python on the NVIDIA Jetson Xavier NX platform. The depth map captured by the Intel RealSense D435i is first transformed into point cloud data using the RealSense SDK. The L-PointPillars algorithm is used to perform object detection on the point cloud data and generate bounding boxes for the detected objects. The bounding boxes are then used to calculate the distance between the obstacles and the BVI individuals.

The system utilizes the Text-to-Speech (TTS) method to provide auditory feedback on the distance of obstacles for BVI individuals. According to the different modes of interaction, existing visual impairment aids can be divided into

two categories: tactile-based interaction and auditory-based interaction. Haptic-based interaction is typically achieved by utilizing vibrating motors or electrodes to stimulate the skin in areas such as the hands and waist, thereby creating a sense of touch. However, this method is limited to conveying simple command messages. Auditory-based interaction differs from haptic-based interaction in several ways. First, sound can be used to convey additional dimensions of information. In addition, hearing is the only remaining means by which BVI individual can perceive their environment remotely. Therefore, the audio feedback needs to be brief and clear, conveying the message without excessively interfering with their natural listening process. The information about the obstacles includes their names and the distances between them and the visually impaired (VI) user. The algorithm for obstacle detection based on L-PointPillars is as follows.

Algorithm 1 Obstacle Detection

```

engine = pttsx3.init();                                ▷ obtain point cloud data
obstacles = ObstacleDetector(pntcloud);                ▷ detect obstacle using L-PointPillars
obstacles = calDistances(obstacles);                    ▷ calculate the distance between the obstacles and camera
for obstacle  $\in$  obstacles do
    if obstacle.distance > 1.5m then
        obstacles.removeObstacle()
    end if
end for
obstacles = sortByDistance(obstacles)
for obstacle  $\in$  obstacles do
    engine.say(obstacle.class + obstacle.dist+'meters ahead')
end for

```

We trained the network on a server equipped with an E5-2620 CPU, 128GB of memory, and 2 Tesla M40 GPUs. A batch size of 8 is used. The KITTI 3D Benchmark includes point cloud data and image data, encompassing objects commonly found in outdoor scenarios, such as cars, pedestrians, and cyclists. L-PointPillars is trained on the KITTI 3D Benchmark, which consists of 3,712 samples for training and 3,769 samples for validation. We augmented the training data by following SECOND [30]. First, we applied the *random_world_flip* along the x-axis. Then, a *random_world_rotation* is performed within the range of $[-\pi/4, \pi/4]$, followed by a *random_world_scaling* within the range of [0.95, 1.05]. The Adam optimizer is used to train L-PointPillars with an initial learning rate of 0.0003. It also has an exponential decay factor of 0.9 and decays every 10 epochs. A batch size of 8 is used. The network is trained on both GPUs for 150 epochs, which took approximately 10 h.

6 Experiment and result analysis

Six volunteers were invited to wear the WETAD and travel through six different outdoor scenes, collecting six point cloud videos. These situations are frequently encountered by individuals with visual impairments during their daily travels. Tactile paving is typically installed on pavements. However, it should be noted that most sidewalks in residential areas do not have tactile paving, which impacts accessibility. Therefore, the sidewalk situation needs to be taken into consideration. The city street scene contains scenarios in which visually impaired individuals are crossing the road or walking alongside it. This section introduces the experiment and analyzes its results.

To evaluate the performance of the proposed WETAD system in detecting obstacles for BVI individuals during navigation, an experimental study was conducted. Six sighted volunteers wore WETAD while traversing six common outdoor environments often encountered by

BVI travelers: city streets, residential sidewalks, building entrances, public transportation platforms, hallways, and open spaces. These uncontrolled environments presented mobility challenges that exceeded the capabilities of the current tactile infrastructure.

6.1 Datasets

KITTI We used the KITTI dataset to train and validate the L-PointPillar algorithm. The KITTI dataset consists of 7,481 training images and 7,518 testing images, along with their corresponding point cloud data. It includes a total of 80,256 annotated objects, with each sample containing a maximum of 15 annotated vehicles and 30 annotated pedestrians. The annotated object categories in the KITTI dataset include *Car*, *Van*, *Truck*, *Pedestrian*, *Person_sitting*, *Cyclist*, *Tram*, *Misc*, and *DontCare*. Each object is also labeled with the truncation level (a floating-point number between 0 and 1 representing the extent of object truncation), occlusion level (0=fully visible, 1=partly occluded, 2=largely occluded, 3=unknown), angle, 2D box, 3D box dimensions, 3D box center, and confidence score. Additionally, the objects are further classified into three difficulty levels based on their visibility in the samples.

(1) Easy: The maximum height of the 3D frame is 40 pixels, the occlusion level is 0, and the truncation level is a maximum of 0.15.

(2) Moderate: The maximum height of the 3D box is 25 pixels, with occlusion levels of 0 or 1, and a maximum truncation level of 0.3.

(3) Hard: The maximum height of the 3D box is 25 pixels, the occlusion level is 2, and the maximum truncation level is 0.5.

In this paper, we use only LiDAR data to train and validate our model. Following the commonly applied convention, the KITTI training set is divided into 3712 and 3769 samples. For training and validation, respectively.

Real-life datasets: We then tests on the obstacle detection of the WETAD device in six different real-life scenarios. Based on the analysis of the characteristics of travel scenarios for visually impaired individuals in section 3.1, these scenarios can be divided into two categories: sidewalk or pathway scenes and road scenes. In this paper, we tested the system in two different sidewalk scenes, two different pathway scenes, and two different road scenes. The tests were conducted online, with participants simulating the daily travel of visually impaired individuals in the six scenarios while wearing the Intel RealSense D435i camera to collect point cloud data. Subsequently, obstacle detection algorithms were run in the laboratory to process the collected point cloud data. To ensure a sufficient number



Fig. 7 Example of how the WETAD is worn

Table 1 Test result of obstacle detection system

Scene	Number of obstacles	Obstacles detected	Detection success rate (%)	Average processing time (ms)
Tactile paving 1	95	90	95	112
Tactile paving 2	103	99	96	125
Sidewalk 1	112	108	96	143
Sidewalk 2	127	122	96	139
City street 1	159	151	95	162
City street 2	184	173	94	184
Average	—	—	95.3	144

of obstacles were captured in each scenario, the data collection time ranged from 20 to 35 min for each scene.

6.2 Experiment settings

The KITTI dataset is divided into 3712 training samples and 3769 validation samples. Three methods are used for data augmentation: random flipping along the x-axis, random rotation within the range of $[-\pi/4, \pi/4]$, and random scaling within the range of $[0.95, 1.05]$. The training of the obstacle detection network is conducted on a server, using the Adam optimizer with an initial learning rate of 0.0003. The learning rate decay factor is set to 0.9, and it is decayed every 10 epochs. The batch size used during training is set to 8, and a total of 500 epochs are trained.

As shown in Fig. 7, the depth camera is secured to the volunteer's chest, while the rest of the WETAD hardware is placed in a waist bag. The volunteers wore the WETAD and traveled through six different outdoor scenes. It took volunteers 20 to 35 min to travel through each scene, and WETAD collected and saved the captured data frame by frame in order to create a point cloud video of their journey. To ensure a consistent number of obstacles can be collected in each scene, the collection time in each scene ranges from 20 to 35 min. The point cloud data collected is stored frame by frame and named according to the timestamp.

6.3 Main result

We compared L-PointPillars with PointPillars and SECOND on the KITTI 3D detection benchmark using the Jetson Xavier NX. The KITTI dataset categorizes the detected objects into three difficulty levels (easy, moderate, and hard) based on the degree of occlusion and height. Table 1 shows the experimental results. The WETAD achieved an average

successful detection rate of 95.3%. On average, it took about 144 milliseconds to process each frame in the 6 scenes. In conclusion, the obstacle detection system can meet the user's real-time requirements. Table 2 shows the results of the comparison. L-PointPillar is approximately 24% and 2.1 times faster than PointPillars and SECOND, respectively, in terms of detection speed. Due to the fewer parameters of L-PointPillar, the AP of L-PointPillar is lower than that of PointPillars for both Car and Pedestrian in all three difficulty levels. However, the AP of L-PointPillar is higher than that of SECOND for both Car and Pedestrian in all three difficulty levels. In terms of Cyclist, the AP of L-PointPillar is higher than that of PointPillars and Second in all three difficulties. In general, we trade off a relatively small amount of accuracy for a significant increase in inference speed.

6.4 Analysis

The first reason that the L-PointPillars improves detection speed lies in its low computational cost. The backbone of

Table 2 Test result of obstacle detection system

Method	Speed (fps)	Car AP			Pedestrian AP			Cyclist AP		
		Easy	Mod	Hard	Easy	Mod	Hard	Easy	Mod	Hard
SECOND	2.91	83.85	74.56	67.22	51.07	42.56	37.29	70.51	53.85	46.90
PointPillars	4.68	86.46	77.28	74.65	57.75	52.29	47.9	80.04	62.6	59.52
Ours	6.17	82.71	75.91	71.85	53.35	48.23	44.52	82.34	64.05	61.43

Fig. 8 Standard convolution used in PointPillars

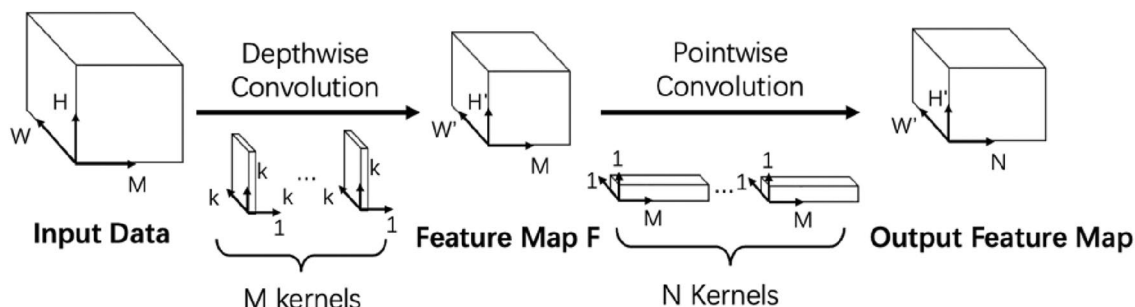
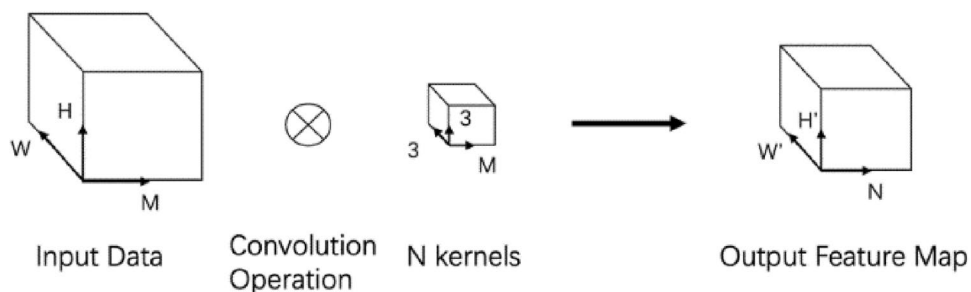


Fig. 9 Depthwise separable convolution used L-PointPillars

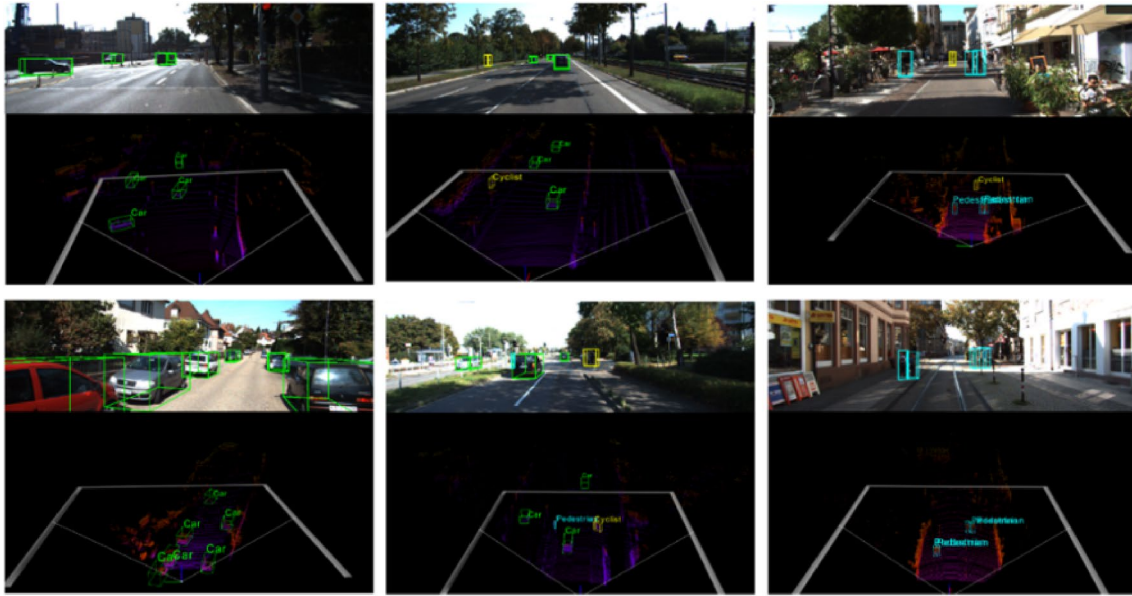


Fig. 10 Visualization of detection result

L-PointPillars uses depthwise separable convolution instead of the standard convolution used in PointPillars, which can reduce computation costs. As shown in Fig. 8, standard convolution employs a single kernel to convolve all input channels. The computational cost of standard convolution with a 3x3 kernel can be calculated according to Eq. 14.

$$Cost_s = W' * H' * 3 * 3 * M * N \quad (14)$$

As shown in Fig. 9, the depthwise separable convolution consists of two parts: the depthwise convolution and the pointwise convolution. In depthwise convolution, each channel of the input corresponds to a specific kernel, and convolution is performed on each channel separately. The computational cost of depthwise convolution can be calculated using Eq. 15. In the pointwise convolution, a 1x1 convolution is performed on the M channels to generate the final feature map. The computational cost of pointwise convolution can be calculated using Eq. 16. The total computing cost of depthwise separable convolution can be calculated using Eq. 17.

$$Cost_{dw} = W' * H' * 3 * 3 * M \quad (15)$$

$$Cost_{pw} = W' * H' * M * N \quad (16)$$

$$\begin{aligned} Cost_{ds} &= Cost_{dw} + Cost_{pw} \\ &= W' * H' * M * (3 * 3 + N) \end{aligned} \quad (17)$$

Based on the above analysis, the ratio of decrease in computing cost between the standard convolution and the depthwise separable convolution can be calculated using Eq. 18, where N represents the number of output channels.

$$R = \frac{Cost_{ds}}{Cost_s} = \frac{1}{9} + \frac{1}{N} \quad (18)$$

Since the output channel of each down-sample block in both backbones is 64, 128, 256, the depthwise separable convolution, which serves as the backbone of L-PointPillars, can reduce the computing cost by almost 9 times compared to the standard convolution used in PointPillars. Depthwise separable convolution, however, doubles the number of convolution layers while reducing the computational cost. In general, using depthwise separable convolution instead of standard convolution reduces the parameter count in L-PointPillar to one-fourth of that in PointPillar.

Since most BVI individuals are sensitive to the price of travel aid devices, high-performance GPUs (such as the Nvidia GTX 1080 ti) are not suitable for their outdoor scenarios due to their expensive price and high power consumption. The Nvidia Jetson Xavier NX is highly suitable for wearable assistive devices due to its compact size and affordable cost. It features 384 NVIDIA CUDA® Cores, 48 Tensor Cores, and 6 Carmel ARM CPUs. Furthermore, its combination with over 59.7GB/s of memory bandwidth makes it ideal for wearable devices that can simultaneously run multiple modern neural networks in parallel and process high-resolution data from multiple sensors. That

is the reason we selected it to develop the WATED and the experiment proves that the WATED has high performance in obstacle detection, as shown in Fig. 10.

7 Conclusion

In this paper, we will first model the obstacles faced by individuals with visual impairments when traveling outdoors. We establish a mapping relationship between the obstacles in the information space and the physical space using point cloud technology. The L-PointPillars network is developed using depthwise separable convolution and an attention model. This network is designed for rapid obstacle detection and is suitable for obstacle detection applications on wearable devices. Lastly, a wearable electronic travel aid device (known as WETAD) based on L-PointPillars is implemented on the Jetson Xavier NX. Experiments show that WETAD achieves a detection accuracy of 95.3%, processing individual frames in an average of 144 ms.

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Data availability The data that support the findings of this study are available in KITTI at <https://www.cvlibs.net/datasets/kitti/index.php> (Geiger et al. 2012).

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

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